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Eddy-Current Method for Measuring the Resistivity of Metals

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A method for measuring the resistivity of metallic specimens is described. The measurement is made by noting the rate of decay of flux from a bar situated in an external magnetic field that has been rapidly reduced to zero. The method is suitable for specimens greater than 5×10^{-3} cm in diameter. For a specimen 1 cm in diameter, resistivities from 10^{-11} to 10^{-3} ohm-cm may be measured with an error of less than three percent. The method requires no contact to the specimen, and local values of resistivity may be measured. Several applications are described.

INTRODUCTION

IF a magnetic field is suddenly applied to a conductor, eddy currents are caused to flow in the conductor. The magnetic field created by these currents is in opposition to the externally applied field, and hence the average field in the conductor approaches, more or less slowly, the value of the applied field. This state of affairs may be made quantitative by noting that by simple manipulation of Maxwell's equations (with the assumption of an isotropic resistivity and permeability and neglect of displacement currents), the rate of change of the flux density is governed by a diffusion-like equation,

$$(\partial \bar{B} / \partial t) = (10^9 \rho / 4\pi\mu) \nabla^2 \bar{B}. \quad (1)$$

The derivation of this equation is set down in many places.¹ In this equation, $\bar{B}$ is the flux density, while ρ is the resistivity in ohm-cm, and μ is the permeability. The magnetic diffusivity, $10^9 \rho / 4\pi\mu$, in a weakly magnetic material ($\mu \approx 1$), is controlled only by the resistivity and can be used as a measure of the resistivity. This diffusivity can be measured by the arrangement shown in Fig. 1. The sample, a rod of circular cross section, is wound locally with a small secondary

(pickup) coil and the whole inserted in a larger magnetizing solenoid. A step function of current is applied through the magnetizing coil, and the resulting voltage across the pickup coil is noted as a function of time. This voltage is caused by the penetration of flux into the rod and so can be used to measure the diffusivity and hence the resistivity.

There are several advantages of this type of technique over the current-voltage method conventionally employed for resistivity measurement. No electrical connections are made to the sample. The sample need not be formed to any particular size or shape. In fact, thick specimens are somewhat to be preferred over wires. Ingots of brittle materials, molten columns, or zone-refined bars may be used directly. Lastly, quite local values of resistivity may be obtained since the pickup coil is mainly sensitive to the material it encloses. This last feature makes possible detection of spatial variations in resistivity.

THEORY

There exist extensive tabulations² of solutions of the scalar equivalent of Eq. (1), since its solution for various boundary conditions is a central problem in the theory of diffusion and heat conduction.

We restrict ourselves, for the moment, to bodies of constant cross section, and stipulate that the magnetic field is applied along a z -directed axis perpendicular to this cross section. Equation (1) becomes a scalar equation for B_z . From a solution of the diffusion equation appropriate to the cross-sectional shape and for a suddenly applied external field, H_0 , we may deduce the function $B(x, y, t, H_0)$ that expresses the flux density as a function of space and time within the specimen. The voltage in the secondary coil will be proportional to the time rate of change of average flux within the material, or exactly

$$V(t) = N \int dx \int dy \dot{B}(x, y, t) \times 10^{-8} \quad (2)$$

where V is measured in volts, and N is the number of turns in the secondary coil.

The appropriate integration for a circular cross

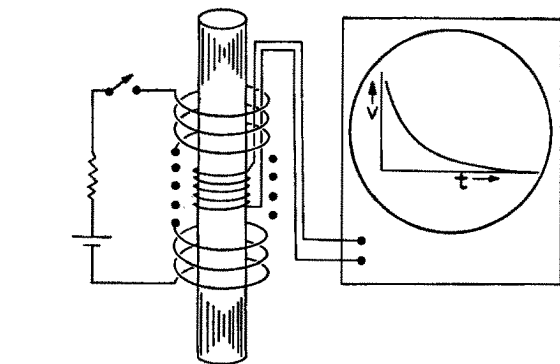


FIG. 1. Arrangement for determining the resistivity of a metal bar. A step-function current applied through the solenoid produces a magnetic field that diffuses into the bar. The resulting voltage across a secondary coil is observed as a function of time on an oscilloscope.

¹ E.g., W. R. Smythe, *Static and Dynamic Electricity* (McGraw-Hill Book Company, Inc., New York, 1950), second edition, p. 390.

² E.g., H. S. Carslaw and J. C. Jaeger, *Conduction of Heat in Solids* (Oxford University Press, London, 1959), second edition.

section of radius R has been performed some time ago by Wwedensky³ and gives,

$$\int ds \dot{B}(r, t) = 10^9 \rho H_0 \sum_{n=1}^{\infty} \exp(-\lambda_n^2 \alpha t) \quad (3)$$

where

$$\alpha = 10^9 \rho / 4\pi\mu R^2$$

and

$$\lambda_1 = 2.405, \quad \lambda_2 = 5.520, \quad \lambda_3 = 8.654, \quad \text{etc.}$$

The λ 's are successive roots of the zero-order Bessel function. While the summation in Eq. (3) may look somewhat forbidding, it is easily seen that at long times only the first term remains, and the voltage approaches a simple exponential

$$V(t) \rightarrow 10N\rho H_0 \exp(-t/\tau_R) \quad (4)$$

where $\tau_R = 2.17\mu R^2 10^{-9}/\rho$. The course of the voltage as a function of time, as deduced from Eq. (3), is plotted as the solid curve in Fig. 2. After a time equal to τ_R , it becomes very close to the simple exponential relationship of Eq. (4).

This feature of the voltage becoming exponential in time is a general feature of all solutions for uniform cross sections. One useful example, discussed in the Appendix, is a rectangular cross section of sides a and b . In the limit of long times, the voltage across the pickup coil becomes

$$V \rightarrow \frac{160N\rho H_0 (a^2 + b^2)}{\pi^3 ab} \exp(-t/\tau_{ab}) \quad (5)$$

where

$$\tau_{ab} = \frac{1.27\mu 10^{-9} a^2 b^2}{\rho a^2 + b^2}.$$

The above results may be generalized to cross sections that are not uniform. The voltage signals may not come to simple exponentials in the time of experiment. However, useful information on the ratio of resistivity at various temperatures may be extracted from the data. This is done by noting that, for an arbitrary shape of specimen, the voltage *versus* time curves at two temperatures have themselves a similar shape. In particular, they become identical if the voltage and time dependences are scaled by the ratio of resistances, as suggested in Fig. 2. Thus, determination of the scaling factor gives directly the resistivity ratio.

EXPERIMENTAL ARRANGEMENTS

The elements of the experimental arrangement, shown in Fig. 1, are a power source, switch, primary coil, secondary coil, and provision for amplification and display of the voltage from the secondary coil. For thin specimens, we use a conventional pulser, while for thick specimens we employ dry cells and mercury or micro-switches. We pass the current through the primary coil

for a time sufficient for essentially complete flux penetration of the specimen, then interrupt this current and note the voltage from the secondary coil caused by the emergence of the flux from the specimen. This procedure has two advantages over the complementary method of watching the flux penetration. First, the current falls faster than it rises because, when interrupted, the series resistance is essentially infinite, and consequently, the inductive relaxation time is essentially zero. Secondly, the external field is accurately zero after the switch is open, while when the switch is closed it may vary slightly due to polarization of the batteries. The design of the primary coil presents no especial problem, except that if the coil is immersed in liquid helium, as we operated, care must be taken to use thick enough wire that the evaporation is not excessive. In our experiments, much less than one cc of helium was vaporized for each observation. The secondary coil is designed by use of Eq. (4) to have sufficient turns to give measurable voltages. The only precautions are to keep the inductance and distributed capacity of this coil low enough that its resonant frequency is much higher than the reciprocal of the relaxation time (τ_R) of the flux and to have sufficient circuit resistance to damp out the ringing oscillations in times comparable to τ_R . For display purposes, we employ a Tektronix

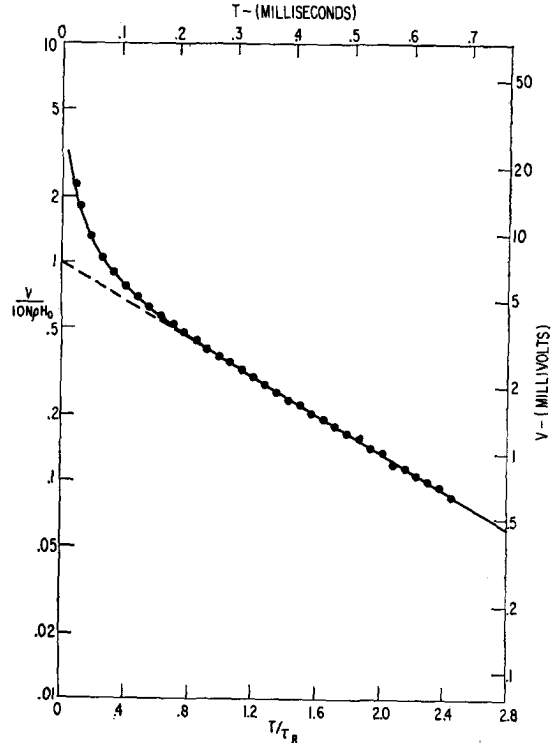


FIG. 2. Solid line: Theoretical curve of voltage across an N -turn secondary coil *versus* time after applying magnetic field H_0 to a cylindrical rod of resistivity ρ . τ_R is a normalizing relaxation time. Circles: Experimental points for a 0.940 cm diam copper rod at 25°C ($N=10$). The corresponding coordinate scales are at the top and to the right.

³ B. Wwedensky, Ann. Physik **64**, 609 (1921).

545 oscilloscope. We calibrate the time scale with a time-mark generator. For specimens of moderately high resistance, we use the dc amplification afforded by the 53/54D plug-in unit for the oscilloscope. For thick, low resistivity specimens, such as the bar of aluminum recorded in Fig. 4, we use a 70-cycle chopper amplifier. By shorting the input for a brief period, we avoid overloading this amplifier. It would be possible in some instances to use a galvanometer directly. The voltage traces are photographed with a Polaroid Land camera and analyzed under a low-power microscope.

EXPERIMENTAL RESULTS

Figure 2 shows the results of an experiment on a rod of AS&R (American Smelting and Refining Company) copper. The experiment is done at room temperature on a bar cut from a coil supplied by the manufacturer. The experimental results are seen to be in close accord with the theoretical predictions if a resistance of 1.76 micro-ohm-cm is assumed. The corresponding value for the resistivity of pure copper at 25°C is 1.72 micro-ohm-cm.⁴ Figure 3 shows measurements on the same sample over a wider range of voltage. This same sample was measured at 4.2°K. The observed resistivity ratio was $\rho(298^\circ\text{K})/\rho(4.2^\circ\text{K})=970$. J. W. Corbett has meas-

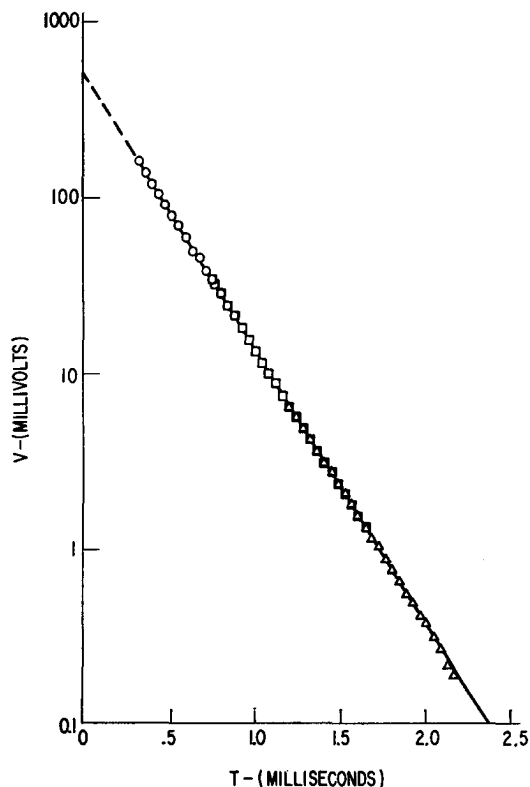


FIG. 3. Further experimental points for the copper rod of Fig. 2 at 25°C ($N=300$).

⁴ A. N. Gerritsen, in *Encyclopedia of Physics*, edited by S. Flügge (Springer-Verlag, Berlin, 1956), Vol. XIX, p. 137.

ured the resistivity ratio by standard techniques of a sample cut from the same coil of copper. He obtained a ratio of $\rho(273^\circ\text{K})/\rho(4.2^\circ\text{K})=938$, equivalent to a room-temperature ratio of 1040.

An application of this technique to zone-refined bars is shown in Fig. 4. The sample is a twelve-pass single crystal of aluminum prepared by K. T. Aust and J. E. Hilliard of this laboratory. At room temperature its resistivity is inferred to be 2.78 micro-ohm-cm. The graph shows the results of an experiment at helium temperature. Here the resistivity is 3.07×10^{-10} ohm-cm. The resulting resistivity ratio of 9.0×10^3 (equivalent to a ratio of 8.1×10^3 if the calculated resistivity at

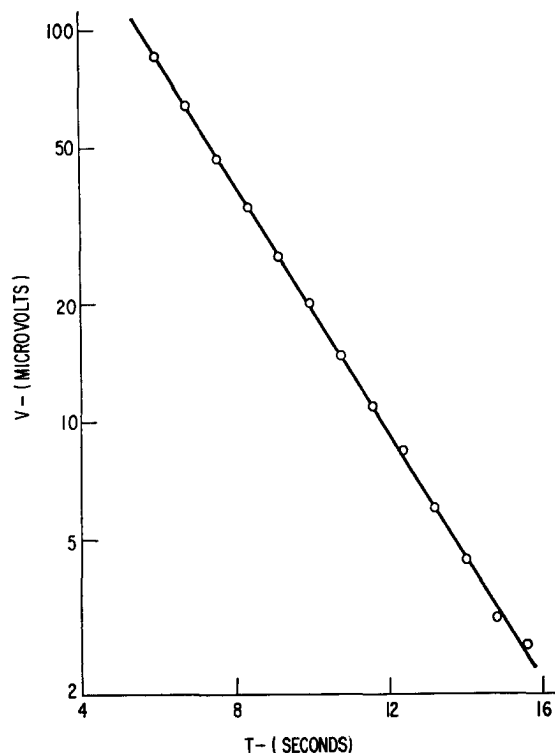


FIG. 4. Experimental voltage-decay curve for an aluminum bar, 1.1 cm $\times$ 1.2 cm cross section, at 4.2°K ($N=2000$). $R_{298^\circ\text{K}}/R_{4.2^\circ\text{K}}=9.0 \times 10^3$. (Zone refined by K. T. Aust and J. E. Hilliard.)

273°K is used) may be the largest yet recorded for aluminum.^{5,6}

The feature of local resistivity measurement is shown in Fig. 5. The sample is a 0.292 cm diameter bar of tungsten that has been given two partially complete passes by H. Schadler, using an electron-beam floating-zone technique. The approximate position of the start of each pass is indicated by the arrow on the horizontal scale. The zone was passed from left to right in this representation. The helium-temperature resistivity measurements in Fig. 5 show that a main purification effect is given by melting in a vacuum, while a sub-

⁵ J. E. Kunzler and J. H. Wernick, *Trans. Am. Inst. Mining Met. Petrol. Engrs.* **212**, 856 (1958).

⁶ M. Caron, *Bull. inst. intern. froid Annexe*, Part 2, 51 (1956).

secondary effect of zone refining is seen in the minimum at about 7 cm on the abscissa.

DISCUSSION

In view of the simplicity and advantages of this eddy-current method, for measuring the resistivity of pure metals, we have been surprised to find no reference to it in the literature. There have been some reports of measurement, by alternating current methods, of the impedance of a coil containing a metallic bar.^{7,8} The portion of the impedance due to the sample may be related to the resistivity of the sample. An alternating-current method more closely related to ours is described by Rorschach and Herlin⁹ who measured the mutual inductance of two coils, corresponding to our primary and secondary. These types of methods suffer from the limitation that they measure the resistivity only within a penetration depth. This value may differ from the resistivity of the bulk material due either to the anomalous skin effect or to the presence of an abnormal surface layer. Other methods have employed the torque in rotating fields,^{8,10} and even the mechanical impulse imparted to a conductor by a sudden change of magnetic field.¹¹ A method closely related to ours has been described by Baird and Boyle.¹² They have measured oscillographically the time required for relaxation of induced currents in a toroidal specimen of copper. Their technique, however, requires this special sample geometry and yields only the average resistivity of the whole specimen.

There are several possible complications in the method we have described. An assumption of this method, in common with other methods, is that one may specify a local value of resistivity. In very pure materials, the mean free path of the electrons may be comparable to specimen dimensions. One must then use an analysis based upon the anomalous skin effect. For the purest metals available at present, diameters of several millimeters will suffice to eliminate this possibility. A second complication concerns the heat evolved in the specimen by joule heating. The energy dissipated in the specimen is $\mu H_0^2 / 8\pi$ ergs/cm³ where H_0 is the applied field. At low temperatures this may cause a heating of the specimen. For pure metals, however, the thermal diffusivity at low temperatures is much greater than the magnetic diffusivity so that if one makes measurements at long enough times the sample will be at ambient temperature. We have never noticed a heating effect. A third complication is the possibility that the magnetic field in the sample may, through magnetoresistance, change the resistivity of the sample.

⁷ G. J. van den Berg and L. C. van der Marel, quoted in reference 4.

⁸ G. Grube and H. Speidel, *Z. Elektrochem.* **46**, 233 (1940).

⁹ H. E. Rorschach and M. A. Herlin, *Phys. Rev.* **81**, 467 (1951).

¹⁰ A. R. Regel, *Zhur. Tekh. Fiz.* **18**, 511 (1948).

¹¹ K. G. Ramanathan and J. S. Dhillon, *Nature* **175**, 948 (1955).

¹² D. C. Baird and W. S. Boyle, *J. Appl. Phys.* **24**, 958 (1953).

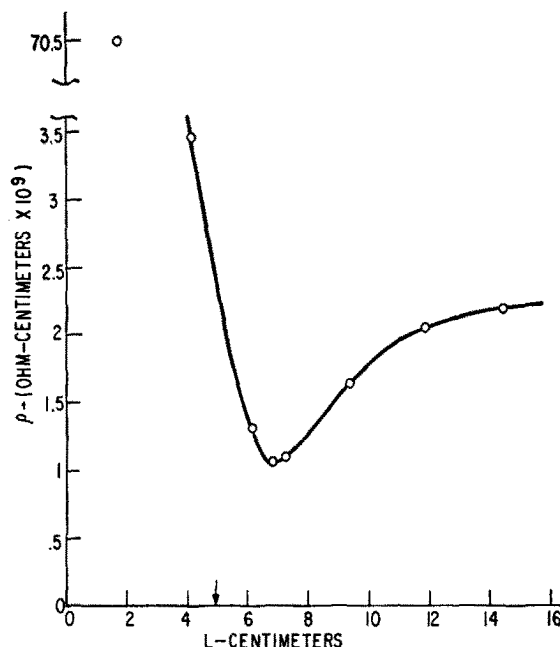


FIG. 5. Resistivity at 4.2°K along a 19.5 cm tungsten rod, refined by the electron-beam floating-zone process (by H. W. Schadler). The zone, swept towards the right, starts at 4.8 cm. The secondary coil was 1.7 cm long and had three thousand turns.

This effect may be detected by observing whether the measurements are independent of field level. We have noticed behavior at high fields in the tungsten specimen that we tentatively ascribe to magnetoresistance. At low fields, however, the measurements are independent of field. Lastly, complications will arise with ferromagnetic or superconducting specimens in zero external field, wherein the assumption of a constant permeability is in error. One may make the measurement in constant external fields that are large enough to saturate the ferromagnet or cause the superconductor to be in the normal state. It is appropriate to recall that the influence of eddy currents on the rate of magnetization of a ferromagnet has been analyzed by Williams, Shockley, and Kittel¹³ and that their influence on the rate of change from the superconducting to normal state of a superconductor has been treated by Pippard,¹⁴ Lifshitz,¹⁵ and Faber.¹⁶

It may be helpful to set down some representative limits of size and resistivity that will restrict the use of this technique. For instance, consider a 1 cm diameter bar. A resistivity of 10^{-3} ohm-cm will give a relaxation time of 0.5×10^{-6} sec. A larger resistivity may present problems in display. The lower bound of resistivity is somewhat more difficult to specify. For resistivities less than 10^{-11} ohm-cm, the relaxation time is greater than 50 sec, and detection mechanisms alternate to voltage

¹³ Williams, Shockley, and Kittel, *Phys. Rev.* **80**, 1090 (1950).

¹⁴ A. B. Pippard, *Phil. Mag.* **41**, 243 (1950).

¹⁵ E. M. Lifshitz, *Zhur. Eksptl. i Teoret. Fiz.* **9**, 834 (1950).

¹⁶ T. E. Faber, *Proc. Roy. Soc. (London)* **A219**, 75 (1953).

display are more easily employed. For instance, one can measure the flux in the bar by moving the sample in and out of a pickup coil that is connected to a standard flux meter. For a specimen 10^{-2} cm in diameter, one is limited to measurement of a resistivity of 10^{-7} ohm-cm or less, inasmuch as this resistivity will give a 0.5×10^{-6} sec relaxation time.

ACKNOWLEDGMENTS

We wish to thank J. W. Corbett and R. M. Walker for emphasizing the application of this technique to zone-refined specimens. K. T. Aust, J. E. Hilliard, and H. W. Schadler kindly made available various specimens as noted in the text.

APPENDIX

We calculate the decay of flux from a bar of rectangular cross section. Consider a bar of rectangular cross section with sides a and b and of infinite length. The origin of coordinates is taken to be the center of the cross section. The solution of Eq. (1) appropriate to these spatial boundary conditions if a field H_0 is removed at zero time is¹⁷

$$B(x, y, t) = B_0 \left(\sum_{m=0}^{\infty} \frac{(-1)^m}{2m+1} \cos \frac{(2m+1)\pi x}{a} \right) \times \exp(-(2m+1)^2 t / \tau_a) \times \left(\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \cos \frac{(2n+1)\pi y}{b} \right) \times \exp(-(2n+1)^2 t / \tau_b)$$

where $\tau_a \equiv [(4\mu a^2 10^{-9}) / \pi \rho]$, $\tau_b \equiv [(4\mu b^2 10^{-9}) / \pi \rho]$, and $B_0 = \mu H_0$. We integrate the flux density over the area of the bar to obtain

$$\int_{-a/2}^{a/2} dx \int_{-b/2}^{b/2} dy B(x, y, t) = B_0 \frac{64ab}{\pi^4} \left(\sum_{m=0}^{\infty} \frac{1}{(2m+1)^2} \exp(-(2m+1)^2 t / \tau_a) \right) \times \left(\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \exp(-(2n+1)^2 t / \tau_b) \right).$$

¹⁷ Reference 2, p. 173.

Differentiating with respect to time yields

$$\int_{-a/2}^{a/2} dx \int_{-b/2}^{b/2} dy \dot{B}(x, y, t) = -B_0 \frac{64ab}{\pi^4} \left[\left(\sum_{m=0}^{\infty} \frac{1}{\tau_a} \exp(-(2m+1)^2 t / \tau_a) \right) \times \left(\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \exp(-(2n+1)^2 t / \tau_b) \right) + \left(\sum_{m=0}^{\infty} \frac{1}{(2m+1)^2} \exp(-(2m+1)^2 t / \tau_a) \right) \times \left(\sum_{n=0}^{\infty} \frac{1}{\tau_b} \exp(-(2n+1)^2 t / \tau_b) \right) \right]. \quad (1')$$

In the limit of very long times, ($t \gg \tau_a$, $t \gg \tau_b$), we may employ only the leading term of each series and obtain

$$\int ds \dot{B} \rightarrow -B_0 \frac{64ab}{\pi^4} \left(\frac{1}{\tau_a} + \frac{1}{\tau_b} \right) \exp \left(-t \left(\frac{1}{\tau_a} + \frac{1}{\tau_b} \right) \right) \rightarrow -(16/\pi^2) 10^9 \rho H_0 ((a^2 + b^2) / ab) \times \exp(-t / \tau_{ab}) \quad (2')$$

where $\tau_{ab} \equiv [(4\mu 10^{-9}) / \pi \rho] [(a^2 b^2) / (a^2 + b^2)]$. The combination of Eq. (2') and Eq. (2) of the text yields Eq. (5) of the text.

Another limit of interest is that of a thin plate ($b \gg a$) at moderately long times ($\tau_b \gg t \gg \tau_a$). Equation (1') then becomes

$$\int ds \dot{B} \rightarrow (8/\pi^3) 10^9 \rho H_0 (b/a) \exp(-t / \tau_a)$$

and the corresponding voltage becomes

$$V \rightarrow (80/\pi^3) N \rho H_0 (b/a) \exp(-t / \tau_a).$$

These last results agree with those deduced by MacColl.¹⁸

¹⁸ L. A. MacColl, quoted by R. M. Bozorth, in *Ferromagnetism* (D. Van Nostrand Company, Inc., Princeton, New Jersey, 1951), p. 784.